Best Practices for Biomedical Research Data Management

Module 3: Contextual Details Metadata

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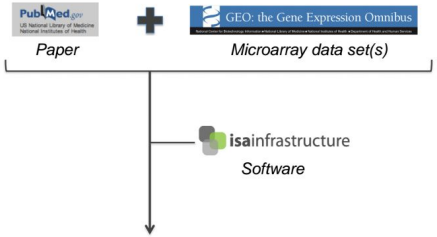

This video will review a retroactive approach to creating metadata and understand issues involved with not creating it more pro-actively. While there may be ways to automatically create metadata pro-actively (ie, prior to publishing the results), there are also ways to create it retroactively based on keywords in the publication. Having researchers create metadata (or co-create it with librarian assistance) pro-actively is generally recommended.

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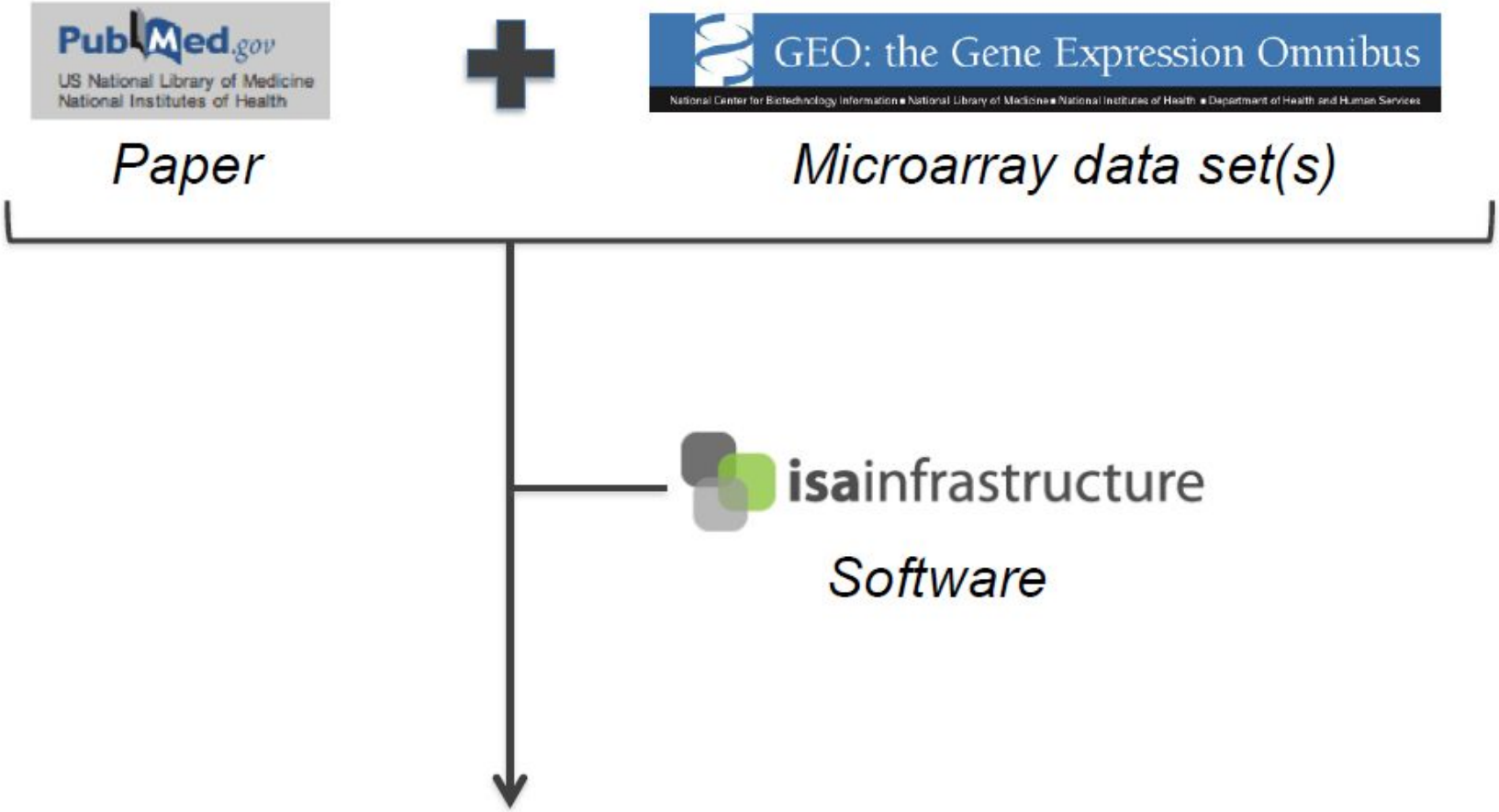
Annotating data sets associated with published papers.


PubMed.gov
US National Library of Medicine
National Institutes of Health
GEO: the Gene Expression Omnibus
National Center for Biotechnology Information ■ National Library of Medicine ■ National Institutes of Health ■ Department of Health and Human Services
Paper
Microarray data set(s)
isainfrastructure
Software

Here's what comes out.

studyinformation

Study ID: ORKINS-1

Title: DNA Methyltransferase 1 is essential for and uniquely regulates hematopoietic stem and progenitor cells

Organism(s): Mus musculus (Mouse)

Description: To examine the role of Dnmt1 in adult hematopoietic stem cells (HSCs), we conditionally disrupted Dnmt1 in the hematopoietic system. Dnmt1 was conditionally deleted by injections of poly(I)-poly(C) to induce Cre expression from the Mx-Cre transgene. Control mice were also injected with poly(I)-poly(C) but do not carry the Mx-Cre transgene. Four days after completion of poly(I)-poly(C) injections, bone marrow was harvested from the mice, antibody-mediated magnetic bead selection was used to remove cells expressing mature lineage markers, and the resulting lineage-depleted cells were stained with fluorochrome-conjugated antibodies against the surface receptors c-Kit, Sca-1 and CD34. Populations of LT-HSCs, MPPs and myeloid progenitors were FACS sorted, RNA was extracted and amplified from these sorted populations and hybridized to Affymetrix microarray chips to compare changes in gene expression induced by conditional knockout of Dnmt1 compared to control in each of the three cell populations. There are 2 biological replicates for the LT-HSCs and MPPs, and 3 biological replicates for the myeloid progenitors. We used microarrays to profile the global gene expression program in hematopoietic stem and progenitor cells following deletion of Dnmt1. DNA methylation is essential for development and in diverse biological processes. The DNA methyltransferase Dnmt1 maintains parental cell methylation patterns on daughter DNA strands in mitotic cells, however, the precise role of Dnmt1 in regulation of quiescent adult stem cells is not known. To examine the role of Dnmt1 in adult hematopoietic stem cells (HSCs), we conditionally disrupted Dnmt1 in the hematopoietic system. Defects were observed in Dnmt1 deficient HSC self- renewal, niche retention, and in the ability of Dnmt1 deficient HSCs to give rise to multilineage hematopoiesis. Loss of Dnmt1 also had specific impact on myeloid progenitor cells, causing enhanced cell cycling and inappropriate expression of mature lineage genes. Dnmt1 regulates distinct patterns of methylation and expression of discrete gene families in long-term HSCs, multipotent and lineage-restricted progenitors, suggesting that Dnmt1 differentially controls these populations. These findings establish a unique and critical role for Dnmt1 in the primitive hematopoietic compartment.

Design(s): perturbation

Publication(s): Trowbridge JJ, Snow JW, Kim J, Orkin SH,DNA methyltransferase is essential for and uniquely regulates hematopoietic and stem progenitor cells.PubMed:19786624

Sample attribute(s):

Attribute name	Attribute value(s)
genetic modification	Dnmt1f/f Cre+ , Dnmt1 conditional deletion by injections of poly(I)-poly(C) to induce Cre expression from the Mx-Cre transgene. Dnmt1f/f Cre- , control, poly(I)-poly(C) injections only, no Mx-Cre transgene
hematopoietic progenitor cell type	hematopoietic progenitor cell SCA1-positive hematopoietic stem cell myeloid progenitor cell
strain	C57BL/6
age	8-12 weeks
organism part	bone marrow
surface protein marker	Lin-c-Kit+Sca-1+CD34+ Lin-c-Kit+Sca-1- Lin-c-Kit+Sca-1+CD34-
label	biotin
phenotype	Defects in HSC self-renewal and differentiation Defects in HSC self-renewal and differentiation on Dnmt1 knockout
organism	Mus musculus (Mouse)

Experimental factor(s):

Factor name	Factor value(s)
genetic modification	Dnmt1f/f Cre+ , Dnmt1 conditional deletion by injections of poly(I)-poly(C) to induce Cre expression from the Mx-Cre transgene. Dnmt1f/f Cre- , control, poly(I)-poly(C) injections only, no Mx-Cre transgene
hematopoietic cell type	myeloid progenitor cell SCA1-positive hematopoietic stem cell hematopoietic progenitor cell

Download:

STUDYMETADATA isatab

more information about the study including protocols applied and sample processing steps... | open isatab in spreadsheet software or download, import and view it in the isacreator

ASSAYDATAFILES&RECORDS

the assays associated with this study are listed below with links to their raw and processed data files (if available) as well as links to submission records in other repositories (where applicable)...

assaytype

Measurement:	transcription profiling
Technology:	DNA microarray
Platform:	Affymetrix

RawData ProcessedData

hematopoietic progenitor cell type
3 recorded

myeloid progenitor cell | SCA1-positive hematopoietic stem cell | hematopoietic progenitor cell

strain
1 recorded

C57BL/6

age
1 recorded

8-12 weeks

organism part
1 recorded

bone marrow

surface protein marker
3 recorded

Lin-c-Kit+Sca-1+CD34+ | Lin-c-Kit+Sca-1- | Lin-c-Kit+Sca-1+CD34-

label
1 recorded

biotin

Where are they now?

- The Stem Cell Discovery Engine is live & available
- So is ISA-Tab software
- Metadata converters developed
- ISA-Tab is now metadata standard for *Scientific Data*

Made by librarians: Northeastern University

Creating a robust, flexible digital repository service.

The vision

A repository robust and flexible enough to support storage, access, & discovery of the widest possible range of digital materials created and used at the university.

Meaning?

Teaching materials	✔
Digital collections	✔
Research data	✔

And more.